

Inside Case

MOTHERBOARD

The **computer motherboard** connects all the parts(components) of a computer. Mainboard, baseboard, mobo (abbreviation), system board, MB (abbreviation), the logic board are the synonyms of a computer's motherboard. The motherboard is the most important component in the PC. All the components such as RAM stick, hard disk drive, optical drives, processor, processor fan, and external card are plugin into the motherboard. **A computer motherboard** is a single platform to connect all of the parts (components) of a computer, Hence it is considered as the backbone of a computer.

Some of the popular manufacturers of the motherboard.

Intel

ASUS

AOpen

ABIT

Biostar

Gigabyte

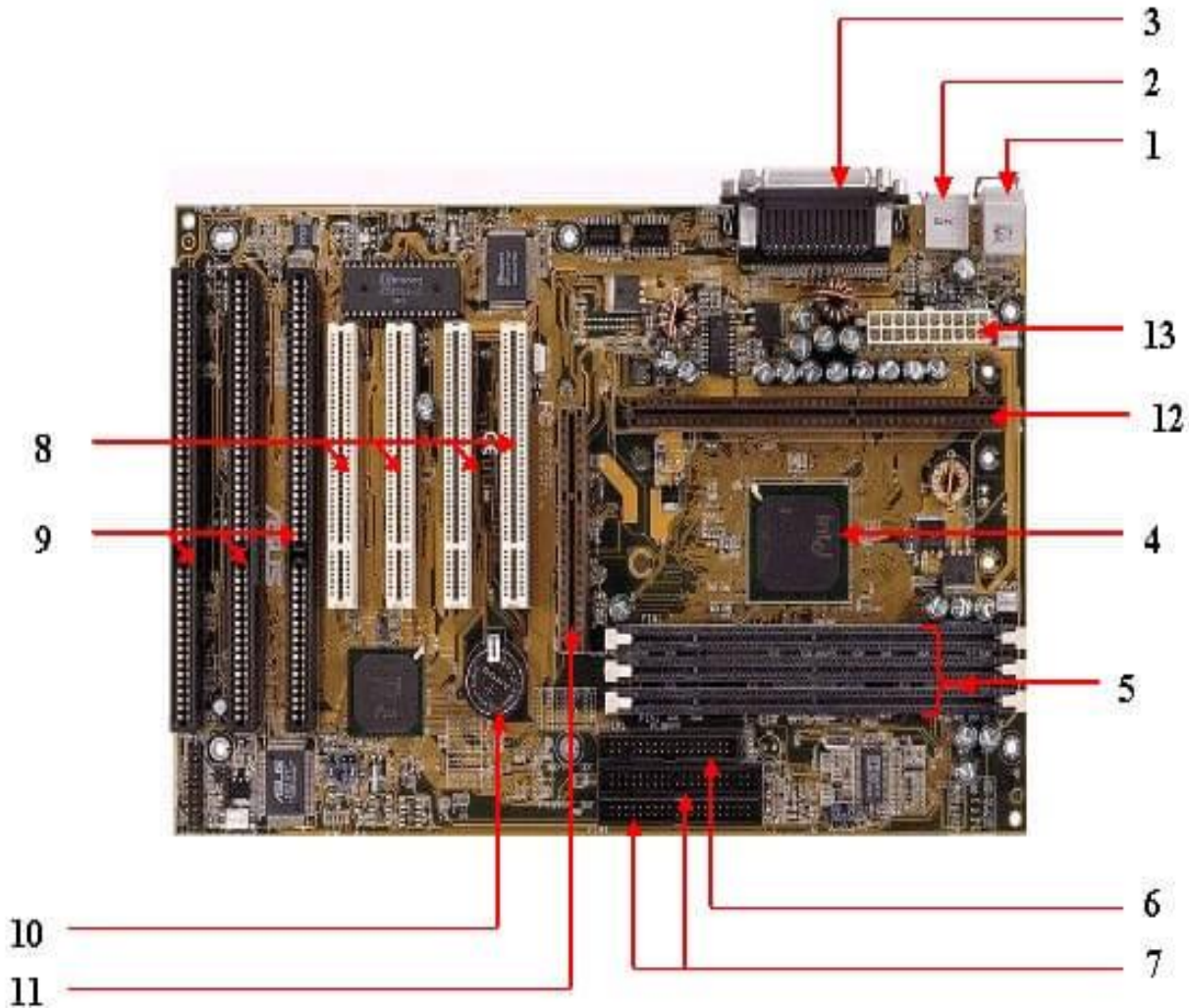
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PORTS AND SLOTS

In the most easy-to-understand terms, "ports" are holes used to plug external devices into your computer. They will be found largely in the back of the computer. "Slots" are holes used to plug internal devices (sound/video cards...etc) INTO your computer. These will be found on the inside of your computer...and the devices you plug into will have "ports" on them...which will be able to be accessed from the back of the computer like the other ones.

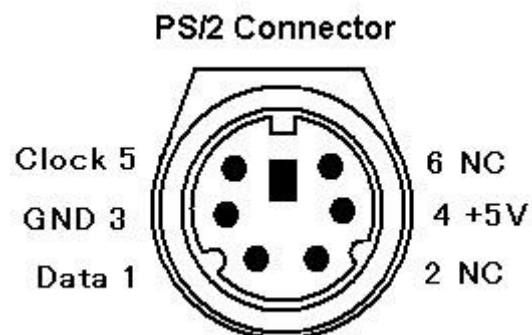
Components of Motherboard

The important components of a Motherboard are given below:



1. mouse & keyboard :

There are two types of keyboard and mouse connectors. The first type is called PS/2 and the second one is called USB.



2. USB (Universal serial bus):

USB is a **Universal serial bus**. It is used for connection for PC. Different devices are used to connect with USB ports such as a mouse, keyboards, scanners, cameras, and even printers. The USB connector is used to connect a computer motherboard and a peripheral device. You can insert or remove peripheral devices connected by a USB connector without restarting your system.

3. Parallel port:

Most old printers are used to connect by the parallel port. Parallel port used more than one wire for sending or receiving multiple bits of data at once, while serial port uses only one wire. Parallel ports use a 25-pin female DB connector.

4. CPU Chip :

CPU refers to a processor, the central processing unit, also called the microprocessor performs all the tasks that take place inside a computer system. It is also known as the brain of the computer.

5. RAM slots :

RAM slots are for attaching RAM on it in general desktop we can see two slots of RAM but in server motherboard, we can see 4+ slots of RAM. RAM comes in different sizes (memory).

6. Floppy controller :

In the old motherboard, the floppy drive connects to the computer via a 34-pin ribbon cable, one end of the ribbon cable is connected to the floppy drive and the other is connected to the motherboard.

7. IDE controller :

IDE is **Integrated Drive Electronics** also called **ATA** or **Parallel ATA (PATA)**. IDE controller is responsible for controlling the hard drive. Today's computers no longer come with an IDE controller.

8. PCI slot :

PCI stands for Peripheral Component Interface, PCI slot allows you to insert expansion cards into your computer. PCI is used to connect additional PCI devices like network cards, sound cards, modems, video cards. Some of today's computers no longer come with a PCI expansion slot. It connects audio, video, and graphics.

9. ISA slot :

ISA stands for Industry Standard Architecture, It is the standard architecture of the Expansion bus. Its connect modem and input devices.

10. CMOS Battery :

CMOS is complementary metal-oxide-semiconductor is used to store BIOS settings in the computer motherboard. CMOS Battery also stores the date and time.

11. AGP slot :

The Accelerated Graphics Port (**AGP**) is a high-speed point-to-point channel for attaching a video card to a computer system. If you have a modern motherboard, you will almost certainly notice a single connector that looks like a PCI slot. A fast port for a graphics card

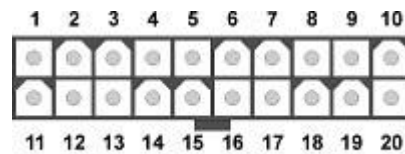
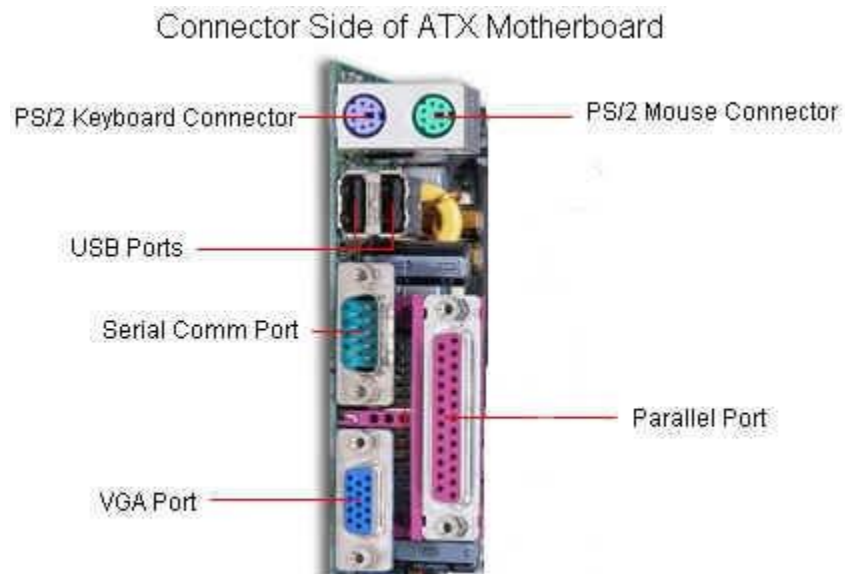
12. CPU slot :

The processor socket (also called a CPU socket) is the connector on the motherboard that connects a CPU.

13. Power supply plugin :

The Power supply provides the necessary electrical power to make the computer system operate. The power supply takes standard 110-V AC power and converts into +/-12-Volt, +/-5-Volt, and 3.3-Volt DC power.

The power supply connector has 20-pins, and the connector can go in only one direction.

**Connector Side of Motherboard**



Keyboard & Mouse: This Port is used to connect the keyboard and mouse, now a day we use a USB connector for the keyboard and mouse



Serial: It is used to connect some types of modem, scanner, or digital camera



Parallel or Printer: You plug your printer into the parallel, or printer, port. But now printers may use a USB port



USB: Designed to replace older Serial and Parallel ports, the USB (Universal Serial Bus) can connect computers with several devices, such as printers, keyboards, mice, scanners, digital cameras, PDAs, and more



Video or Monitor: It used to connect your monitor to the video port



Line Out: It is used to connect speakers or headphones into the Line Out jack



Line In: The Line In jack allows you to listen to your computer using a stereo system



Microphone: It is used to connect a microphone into this jack to record sounds on your computer



Joystick or Game: If you have a joystick, musical (MIDI) keyboard, or another gaming device, this is where you plug it in



Phone or Modem: The phone or modem jack is where you plug your computer into a phone line

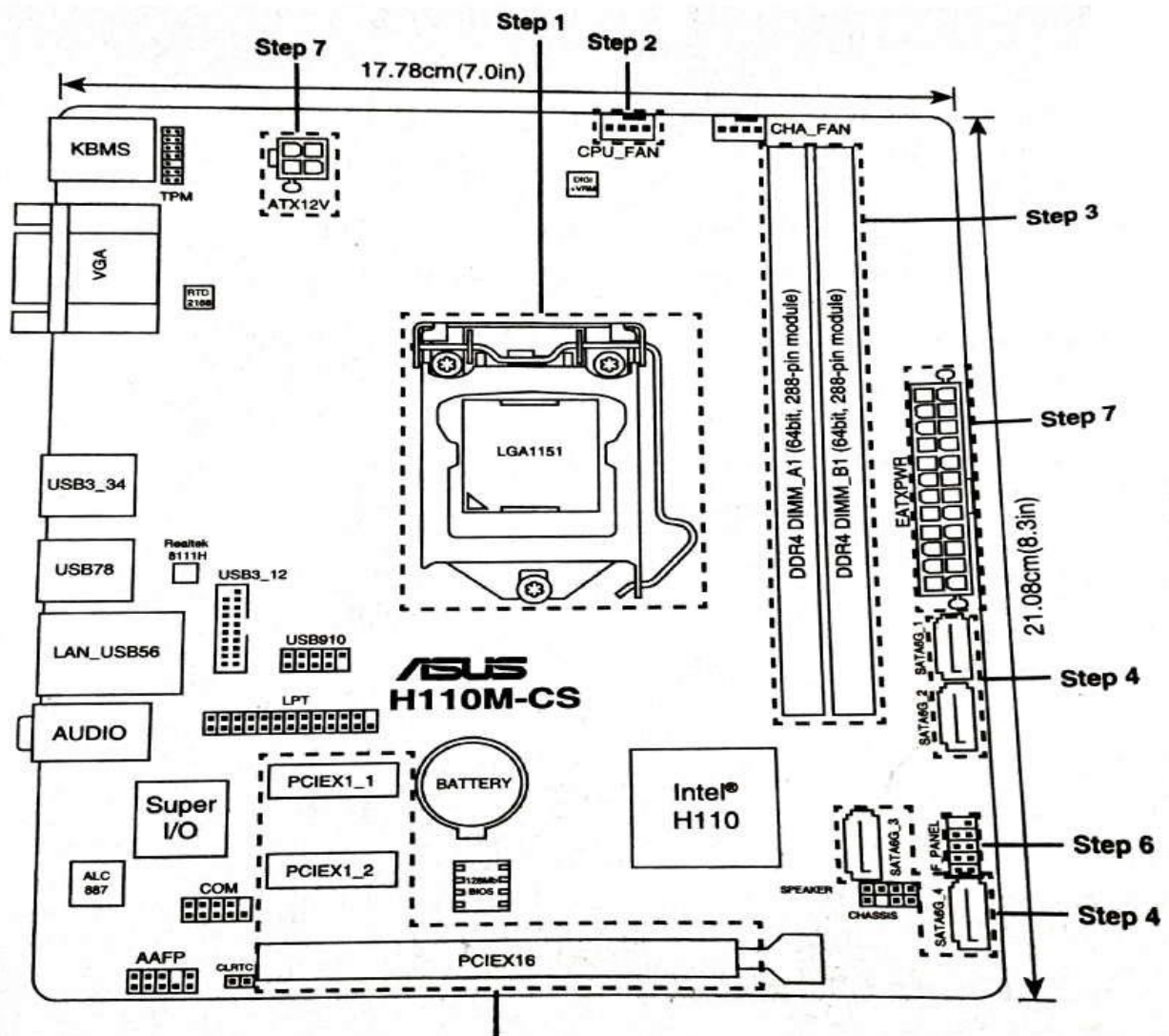


Network or Ethernet: You can connect your computer to a network by plugging in an Ethernet cable in this port

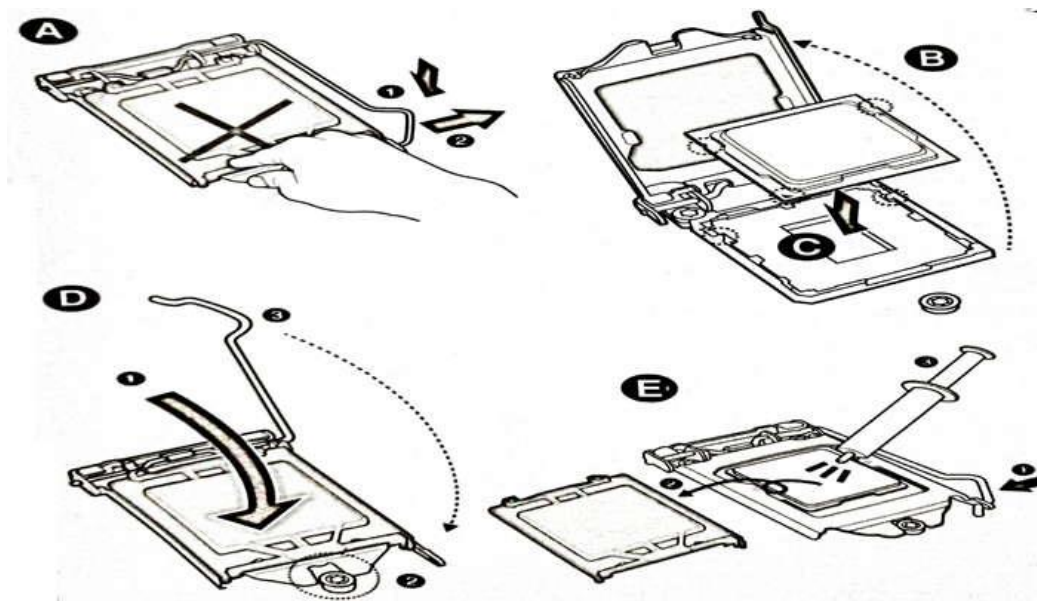


SCSI: It used to connect a hard drive, CD-ROM drive, or another device to a computer

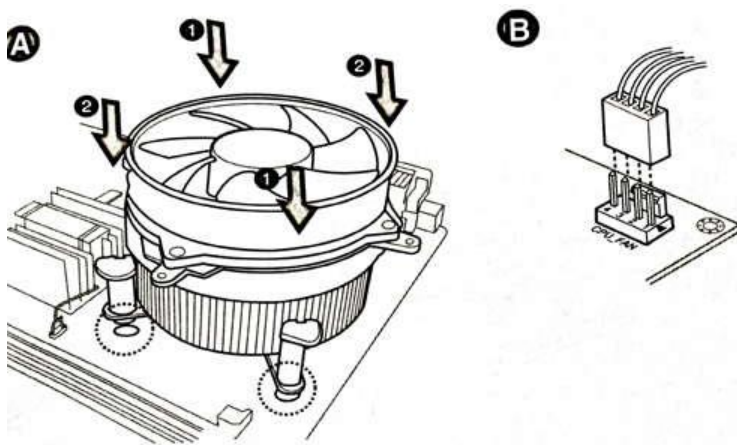
Installing components of the motherboard

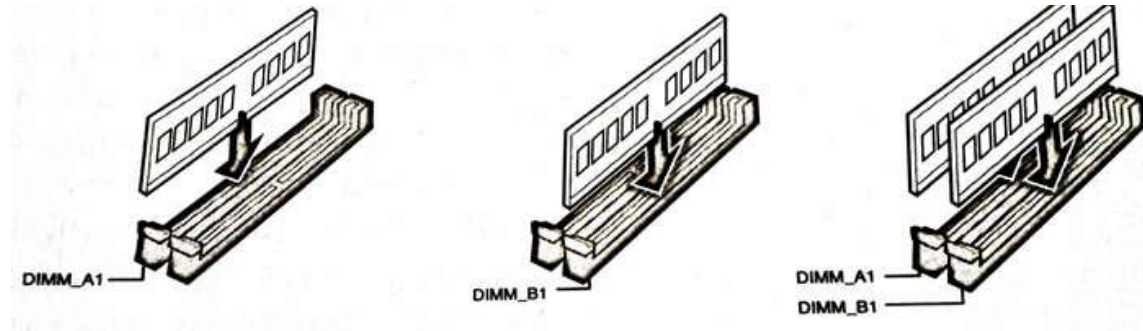
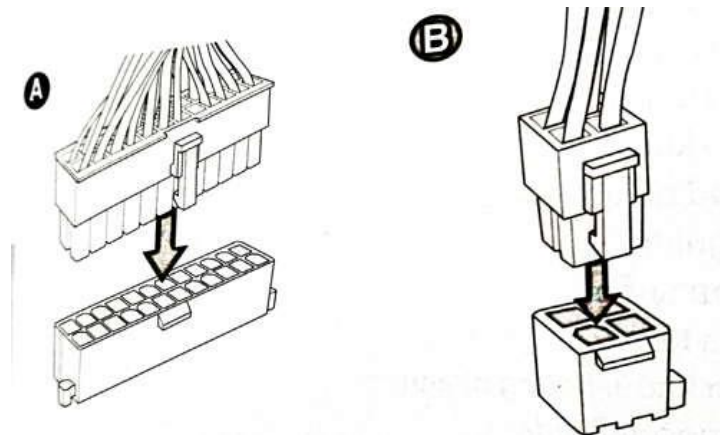


Step 1) Install CPU



Step 2) Installing CPU Fan



Step 3) Install RAM**Step 4) Install SATA devices****Step 5) Install Expansion cards****Step 6) Install system panel connector****Step 7) Install ATX power connector****Step 8) Connecting input/output devices****Step 9) Power On the system and install Operating System****Central Processing Unit**

The processor is a chip or a logical circuit that responds and processes the basic instructions to drive a particular computer. The main functions of the processor are fetching, decoding, executing, and writing back the operations of an instruction. The processor is also called the brain of any system which incorporates computers, laptops, smartphones, embedded systems, etc.

A CPU is further divided into three sections.

- ALU (Arithmetic and Logical Unit)
- MU (Memory Unit)
- CU (Control Unit)

The computer is very much like humans, and the CPU is its brain. We will explain to you the function of the CPU with human examples.

ALU (Arithmetic and Logical Unit):

Arithmetic and Logical Unit performs all the calculations on the computer.

MU (Memory Unit):

The Memory Unit in a CPU is useful to store data. There are several **types of storage devices in a computer**. We have discussed more on ROM and other storage devices in another the above-linked article.

CU (Control Unit):

The Control Unit in a CPU is responsible for giving commands and controlling other parts of the computer.

Example:

If someone asked you to calculate $2485+6576$, your brain will process the data and provide the answer. Likewise, when a user gives any command to the computer, the CPU processes the data and provides the output with the help of ALU.

Similarly, like the human brain stores data and memories, the CPU uses a Memory Unit to store the data.

Moreover, the human brain also controls other parts of our body. Likewise, the brain of the computer (CPU) controls other components of the computer.

Types of Processors in a Computer:

There are different types of Processors like single-core, dual-core, quad-core, octa-core, and multi-core. As the name suggests, a single-core CPU has only one core, a dual-core CPU has two cores, and so on.

- A Single-Core CPU has only one core and is not suitable for performing multiple tasks.
- You can use Dual-Core CPUs if you don't use the computer very often. It is also ideal for students.
- A Quad-Core and Hexa-Core processors are ideal for most users. Octa-Core and above CPUs are suitable for high usage.
- If you are working in an office where you need to perform multi-tasking, you might need Octa-Core or higher.

However, gamers need different strategies while buying a CPU. More cores do not mean it benefits you in gaming. The most important component for gamers is GPU.

Cores:

The size of the CPU chip remained the same, but more cores have been added to it with the help of nano-technology. More cores mean more speed and performance. With multiple cores, you can operate several operations at once. However, some activities like rendering and gaming consume too much CPU usage and slow down the computer. Therefore, engineers introduced GPU, which can perform graphical operations quickly and effortlessly.

Example: Assume the processor is a robot, that has multiple hands (cores) to perform activities (tasks). When you give it a small task, it will perform the task with a single hand, like playing movies, watching pictures, listening to songs, working on MS-word, etc. If you give it 8 smaller tasks, it will perform the tasks with eight hands.

However, if you give it a bigger task, it should use multiple hands to perform that single task like gaming and video processing. Therefore, having an 8 core (octa-core) CPU can perform several smaller tasks or one huge task.

Threads:

Threading is a process where the Core of the CPU is divided into two or more virtual parts so that each part can perform different tasks. CPU manufacturers use treading technology to split the cores virtually and enhance the performance.

Whenever you are threading a dual-core CPU, it behaves like a quad-core CPU. If you have 6 cores with 6 threads, your CPU might run as a 12 core processor.

A simple explanation: When you are performing smaller activities like watching movies, listening to songs, etc., each process takes the entire core. Therefore, threading divides the cores and spread the tasks so that your CPU can handle several tasks at once. If you have a 4 core CPU with 8 threads, it will run equivalent to the 12 core CPU.

4 core CPU with 8 threads = 6 core CPU with 6 threads.

Cores are hardware chips located on the CPU, which cannot be divided physically. Threading is software that divides the cores virtually. However, this technique doesn't work when you are playing games or using designing software.

Clock speed and **Overclocking**:

Clock Speed shows how many instructions your CPU can process in a second. It is represented by GHz and pronounced as GigaHertz. Here

G means Million and Hz mean time per second. For example, a 4 GHz CPU indicates that it can process 4 million instructions per second.

More clock speed = faster reaction time = better performance

Therefore, gamers use the technique of overclocking to enhance the performance of the CPU. However, overclocking would heat up and burn the processor. So, if you are trying to overclock any component of the computer, make sure you enhance the cooling system.

Also, note that all CPUs cannot be overclocked. Your Processor and Motherboard should have the feature. Anyhow, all AMD CPUs support overclocking. Whereas in intel, only the CPUs ending with the letter k supports overclocking.

Transistors:

Each core in the CPU has thousands of transistors. Lately, CPU manufacturers are trying to reduce the size of these transistors. The size of a transistor is measured in nm (nanometer). Previously the transistors were 14 nm as a standard size. But, these days, you will find 7 nm CPUs in the market. The size may further decline in the future, improving the performance of the CPU.

What is Cache Memory (In Computer)?

In simple words, Cache Memory (aka CPU Cache) is another type of storage medium that stores a small amount of temporary data. Usually, Cache Memory is smaller in size but faster than other storage devices. It is pronounced as Cash, and you will find it on both computers and mobile devices.

Basically, there are four **types of storage on your computer**. One is your Hard Disk, another one is RAM, the third one is Registers, and the fourth one is Cache. The Types and Sizes of Cache are further explained in this article.

Simple Example: Suppose you are a CPU(Processor), and the computer is your office building. The storage room (hard disk) has all your data, but it takes a lot of time for you to walk and grab a file from the storage room. Similarly, it takes a long time for a CPU to access the files stored in your Hard Disk Drive.

Now, suppose your office cubicle is RAM. You can store the data, which you regularly use in your cubicle so that you can quickly take the files. Similarly, some data is stored in RAM for quick access. Just like your cubicle has lesser space than your storage room, the RAM has less storage capacity than the hard disk.

Now the Cache is like your desk (cabinet). It is closest to you, you can pick up files without getting up from your seat, but it has very limited space. Likewise, Cache Memory stores only a small amount of data, but it is placed closest to the CPU so that the CPU can quickly access the information.

What is Cache Memory in Android?

The Mobile Cache is similar to the computer Cache, as mobile phones are also a type of minicomputers. The memory used by the CPU to run the applications and files faster on a mobile phone is called Cache Memory.

What is Internet Cache or Browser Cache?

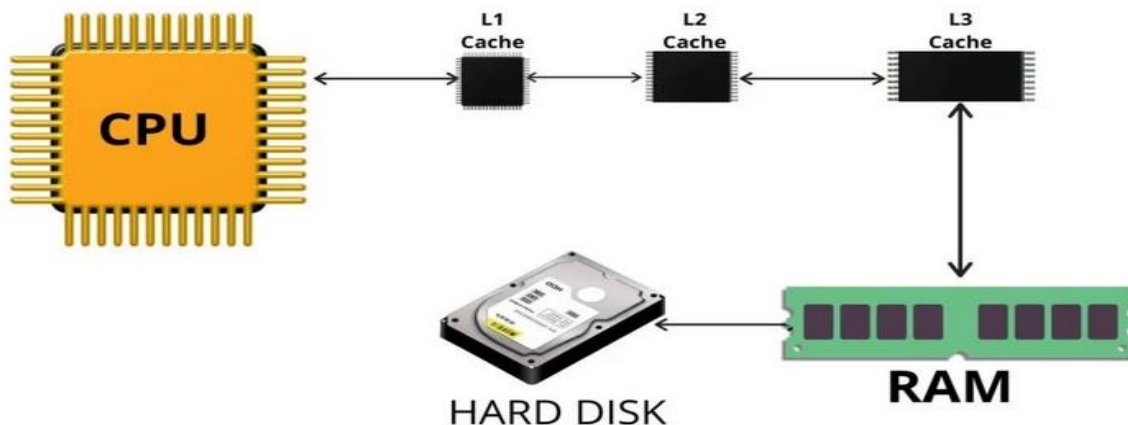
Internet Cache is also known as browser Cache or web Cache. When you visit a new website on your web browser, it takes longer to open. It is because the browser has to load all the elements of the website.

Hence, the web browser saves some of the data on Cache Memory so that it can load the webpages faster. But, if your Cache Memory is full, it might show down your browsing speed when you visit new websites.

Therefore, most users clean/remove the Cached Data frequently. We Provided a step-by-step procedure on how to remove Browser Cache. You can scroll down or use the Table of Content to jump to the segment.

How Does Cache Memory Work? (Functions of Cache Memory)

The only function of Cache Memory is to store information of the recently-used programs. So, whenever you send a request to the Processor (CPU), the CPU first scans the Cache Memory (Level 1, 2, and 3) for that information, then the RAM, and then the hard disk. Therefore, if the data is already stored in Cache Memory, the CPU will quickly process it.



Real-life example: When you open Facebook or any other website in your internet browser, it stores some data in the Cache. So, when you open the same website again, it loads faster than before. It is because some of the data like thumbnail, logo, and images were already present in Cache, and your web browser doesn't have to reload it again.

Another Example: When you open your gallery app, it stores all the folder thumbnails in the Cache Memory. So, whenever you open the gallery app, the mobile processor doesn't have to load the thumbnail images again and again.

How Cache Affects your PC Performance?

- Cache Memory is the fastest memory on the computer.
- The Cache Memory is located inside or near the processor (CPU) for quick access.
- Cache Memory reduces the load on the CPU

As we discussed earlier, when you send any command to the CPU, the CPU first scans the Cache Memory for the required data. Therefore, the CPU can quickly launch the program without scanning RAM or Hard Disk Drive. It is also necessary that you check the Cache Memory before buying a new computer. More Cache Memory stores more data, and the CPU can launch more applications or websites quickly. Cache Memory is pretty expensive, but it also improves the system performance.

Types of Cache Memory:

There are three types of Cache Memory that are Level 1, Level 2, and Level 3.

Level 1 (L1 Cache):

Level 1 Cache Memory is placed inside the CPU core. It sizes around 2KB to 256KB, and it is the fastest type of memory. Since it is located inside the CPU core, the CPU can access it fast.

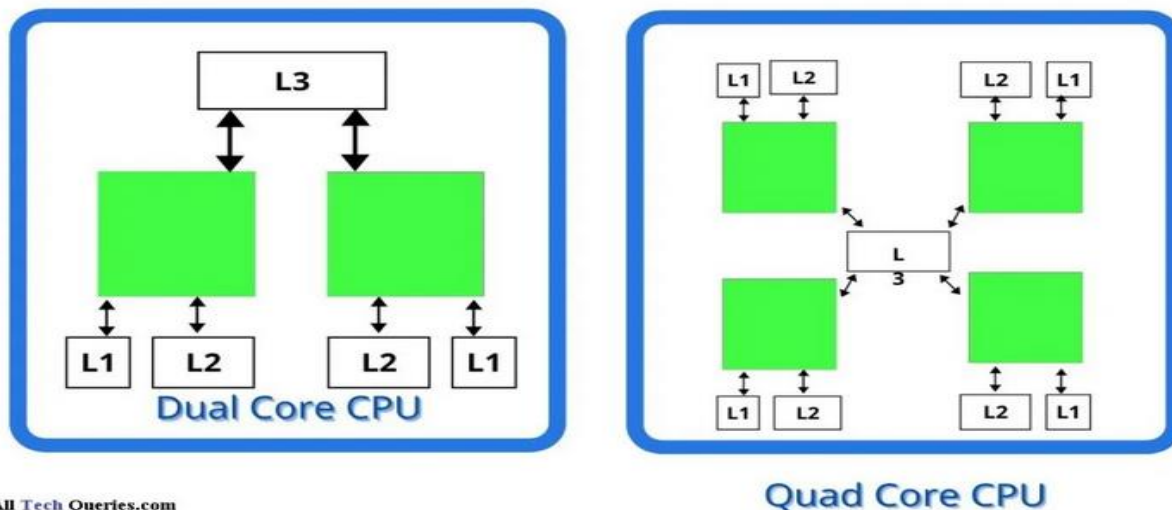
Level 2 (L2 Cache):

Level 2 Cache Memory could be inside the CPU or located on the motherboard near the CPU. If the Level 2 Cache Memory is located on the motherboard, it is connected with a high-speed bus to transfer data quickly. These days, CPU manufacturers are integrating Level 2 Cache inside the processor. Level 2 Cache has more storage but performs slower than Level 1 Cache. It sizes from 256KB to 8MB in general.

Level 3 (L3 Cache):

Level 3 Cache is usually available only in high-end processors. The storage capacity of L3 Cache is around 4MB to 60MB, but it is also the slowest type of Cache Memory. Though, it is very much faster than the RAM and Hard Disk Drive.

These days, you will find multiple cores in a CPU like Dual-Core, Quad-Core, Octa-Core, etc. All the cores of a CPU have their individual/specific L1 and L2 Cache. But, L3 is a common Cache Chip shared between all the cores of a CPU. Moreover, you might also see a Level 4 Cache appearing in the future.



What are CPU Registers?

CPU Registers are temporary storage devices built inside a CPU (Processor). When you give any instructions to the CPU, it stores the instructions or data in the registers till the process is completed. A Register can hold Data, addresses, and Instructions.

Also, Read: *What is a CPU (Processor)? Types of Processor in a Computer.*

Difference Between Cache Memory and Registers:

Cache Memory is also a storage device that holds data temporarily. Hence, most people confuse between these two, we will explain them in simple words with understandable examples.

To explain in simple words, Registers store the data before processing, and Cache Memory stores the data after processing. Check out this example to understand the difference between Cache Memory and Registers easily.

Real-Life Example: Let's take the office example again, which we have discussed in the **Cache Memory** article. As I said, Cache Memory is like the cabinet desks where you store the frequently used data. Now, suppose your boss has given you a new project (instruction). You will keep the files on your working desk, analyze them, then store them in your cabinets or storage.

Similarly, when you give the CPU any instructions, it puts the information in the registers, processes it, then stores it on Cache, RAM, or Hard Disk.

Cache Memory	Registers
Cache Stores Processed data.	Registers store Unprocessed data.
Cache Memory could be located inside the CPU or on the motherboard.	Registers are a part of the CPU.
CPU uses Cache to detect previously stored data.	CPU uses registers to process new data.
Cache Memory Stores recently used data.	Registers store the data that the CPU is currently processing.

Now that you recognize the difference between Register and Cache Memory, let's learn a few more things about Registers.

Types of Registers in Computer:

CPU Registers are classified into 6 types based on their functions.

- Memory Address Register (MAR)
- Memory Buffer Register (MBR)
- Memory Data Register (MDR)
- Instruction Register (IR)
- Program Counter (PC)
- Accumulator

Register Size:

Register size determines how much information the CPU can store it in. A Register can be of sizes 1 Byte, 2 Bytes, 4 Bytes, 8 Bytes, or 16 Bytes. Having a bigger size Register can increase the performance of the CPU.

Which is faster? Cache or Registers:

CPU Registers are faster than CPU Cache. In fact, Registers are the fastest **type of storage device** in a computer. Registers are also the most expensive type of computer memory.

How to see the size of the Register?

Usually, the Register size is shown in Bit. A 32-bit Processor has 32 bits, which means 4 bytes. Similarly, a 64-bit Processor has 64 bits, which is 8 bytes.